Particle detection and tracking in fluorescence time-lapse imaging : a contrario approach

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Detection Theory

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Introduction

Fluorescence microscopy enables clinicians and researcher to view biophysical particles in a sample. This process relies on staining the target particle with a fluorescent chemical so that it can respond to excitation and be sensed.

Nevertheless, raw data takes the form of videos, which cannot easily be handled for interpreting particle-related information in an automatic way for quantitative analysis. Consequently, there is a need to develop tools to automatically and reliably detect and track particles in data obtained by fluorescent microscopy. Dimiccoli et al. [2016] proposed to leverage the a-contrario framework to provide a solution for particle detection and tracking.

This project aims at understanding and reproducing the work of Dimiccoli et al. [2016] into three steps. First, synthetic data will be generated to obtain a baseline on which the method will be evaluated. Second, the particle detection will be analyzed and evaluated. Thirs, the point tracking approach used for particle tracking will be analyzed and evaluated. The code for this project is available at https://github.com/Octave-Jeanne/Detection-Theory-MVA and slides with illustrative animations can be found at https://docs.google.com/presentation/d/1sMnkO4-Ld5RiTI0ma4UCghtQmXJm-AUobeIfptVcEoY/ed

1 Data analysis and generation

1.1 Real-Data obtention

A common way to detect particles in biophysical systems is fluorescence microscopy. This process can be decomposed into two phases :

1. Staining : This phase aims at attaching a fluorescent entity to the target particles by leveraging biochemical processes Im et al. [2018].

2. Observation : In this phase, the media is excited by a well chosen light source so that the chemical that were used to stain the target particle reacts through fluorescence. The re-emitted light is then detected. Since the emission source is attached to the target particle, the area that are occupied by such particle present hight values in the detector's output (More about the process can be found at this website Widefield Fluorescence Microscopy)

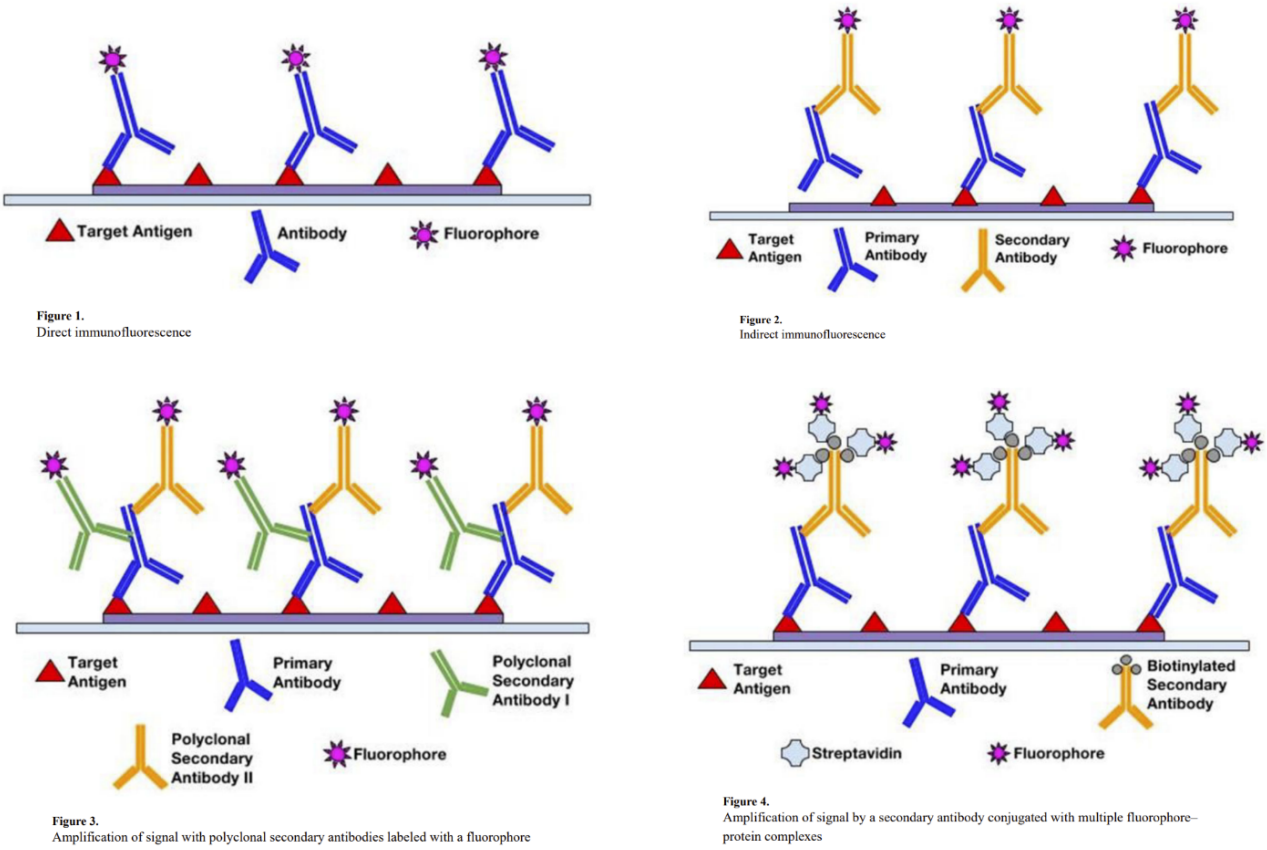

Figure 1: Examples of Staining procedures from Im et al. [2018]

The physics apsects of this process inherently makes it return data that typically resemble figure 3

1.2 Artificial Data Generation

The rest of the project requires to access data resembling real-world cases to evaluate the method that will be proposed. While Icy provides a good quality benchmark to generate synthetic videos of particle viewed through the lens of fluorescence microscopy, ground truth could not be extracted making the benchmark unusable for quantitative analysis.

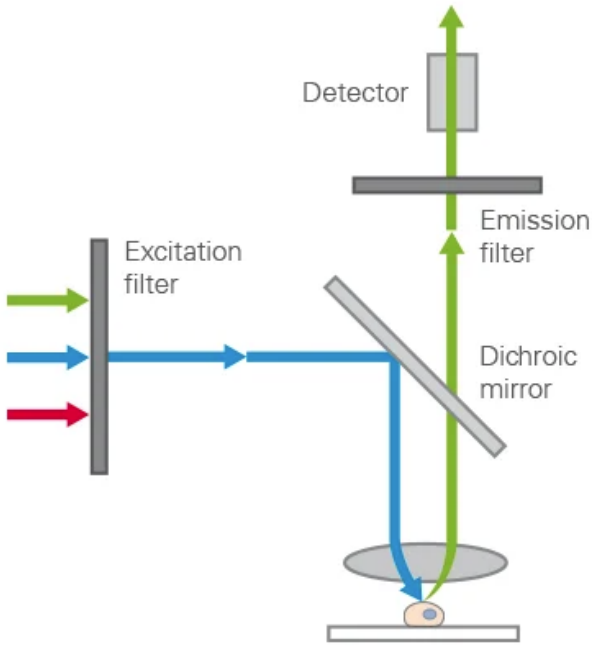

Figure 2: Example of a fluorescence microscopy pipeline from Widefield Fluorescence Microscopy

Figure 3: Example of real-world data from Dimiccoli et al. [2016]

Consequently, data was generated by hand using the following assumptions :

- Each particle follows a Brownian Motion.
- Each particle is represented by a gaussian-looking spot.
- Each frame is subject to a white background noise.

While these assumptions are simplistic, they are also reasonable since real-world data particles actually look gaussian and since the Brownian Motion was historically introduced to model the trajectory of biophysical particles. This statement is also backed by the fact that the handcrafted simulation reasonably matches the output of Icy.

2 A-contrario Particle Detection

2.1 Methods

2.1.1 Overview

The particle detection method that is considered in this project correspond to the one used in Dimiccoli et al. [2016]. This method itself is a particular application of the more general paper Grosjean and Moisan [2009]. It relies on an a-contrario approach that assumes a white noise background and pixel neighborhood gaussianity evaluation measurement statistic.

2.1.2 Background model

When there is nothing to detect, the output of the sensor is just sensor-noise, which is effectively modeled by a stationary gaussian white noise process. This process is defined as a linear transformation of the Schwarz derivative of the Brownian Motion. More formally, a stationnary white noise $W_{\mu,\sigma}$ of mean μ and standard deviation σ is a random distribution satisfying :

$$\langle W_{\mu,\sigma}, \varphi \rangle \sim \mathcal{N}\left(\mu, \sigma \int_{\mathbb{R}^2} \varphi(x)dx\right), \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^2)$$

$$\mathbb{E}[\langle W_{\mu,\sigma}, \varphi \rangle] = \mu, \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^2)$$

$$\mathbb{E}[\langle W_{\mu,\sigma}, \varphi \rangle \langle W_{\mu,\sigma}, \psi \rangle] = \int_{\mathbb{R}^2} \varphi(x)\psi(x)dx, \quad \forall (\varphi, \psi) \in \mathcal{D}^2(\mathbb{R}^2)$$

In the case of the project, each images corresponds to a $H \times W$ grayscale pixel grid. Therefore, each pixel value corresponds to an evaluation of the image process distribution at the test function $\mathbb{I}_{ij}(x) = \mathbb{I}\{i \leq x_0 < i+1; j \leq x_1 < j\}$.

Under the backgound assumption, the image is a stationnary linearly scaled white noise. Therefore, the background model corresponds to a grid of iid random variables following $\mathcal{N}(\mu, \sigma)$.

2.1.3 Measurement Statistic

A qualitative observation of real-world data in figure **??** shows that particles have a disc-like structure where the center area is brighter than the outer-rim and where the brightness smoothly decreases. This visual resembles a plot of a gaussian probability distribution.

Following that observation, the measurement statistic is designed to evaluate whether a local neighborhood present this gaussian-like structure. To do so, one looks at the normalized difference between the average pixel intensity in a centered disc and in its outer-rim as it is illustrated in figure 4.

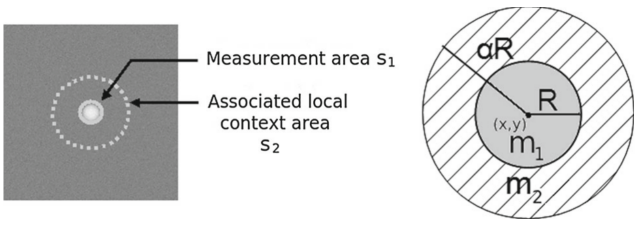

Figure 4: Particle Detector Measurement

More formally for $R > 0$ and $\alpha > 1$ two tunable parameters and I the image :

- The average pixel intensity in the disc centered in x of radius R is measured by m_1, which is defined by

$$m_1(I_{ij}) = (s_1 * I)(I_{ij}), \quad s_1(I_{kl}) = \frac{1}{\pi R^2} \mathbb{I}_{\mathcal{B}(I_{ij},R)}(I_{kl})$$

- The average pixel intensity in the outer-rim of the centered in x of radius R is measured by m_2, which is defined by

$$m_2(I_{ij}) = (s_2 * I)(I_{ij}), \quad s_2(yI_{kl}) = \frac{1}{\pi R^2 (\alpha^2 - 1)} \mathbb{I}_{\mathcal{B}(I_{ij},\alpha \cdot R) \setminus \mathcal{B}(I_{ij},R)}(I_{kl})$$

- The measurement is then the following statistic :

$$m = \frac{m_1 - m_2}{\mathbb{E}\left[\|\langle W_{\mu,\sigma}, s_1 - s_2 \rangle\|_{L_2}\right]}$$

  Where $\mathbb{E}\left[\|\langle W_{\mu,\sigma}, s_1 - s_2 \rangle\|_{L_2}\right]$ is a normalization factor that is used to have $m|\mathcal{H}_0 \sim \mathcal{N}(0,1)$.

In the original paper, the normalization term is noted $\sigma \|\phi_1 - \phi_2\|$, where $\phi_i = s_i * \mathcal{N}(\mu, \sigma)$. However, this term is ambiguous with respect to the norm and the nature of ϕ_1 and ϕ_2, which is why it was replaced in this project.

2.1.4 Number of False Alarms

Following the a-contrario paradigm, the statistical measurement and the background model can be used to define an NFA using the usual construction framework :

$$\text{NFA}(x) = \#\text{Tests} \cdot \mathbb{P}_{\mathcal{H}_0}(T(x) \leq T(X)), \quad \text{Where } T \text{ is a statistic and } x \text{ is a potential realization of } X$$

In the specific case at hand, the neighborhood gaussianity measurement m is designed to satisfy $=m(\mathcal{I}_{ij})|\mathcal{H}_0 \sim \mathcal{N}(0,1)$. Therefore, the standard NFA associated with it writes

$$\text{NFA}(I_{ij}) = \#\mathcal{I} \cdot \Phi(m(I_{ij})), \quad \text{Where } \Phi \text{ is the tail of the normal distribution}$$

By standard construction, this NFA is a well defined number of false alarm.

2.2 Implementation

2.2.1 Reformulation of the NFA

While the NFA's definition is theoretically sound, it requires to compute the tail of the normal distribution at every evaluation, which is computationnally costly. The authors of Dimiccoli et al. [2016] mitigate this flaw by reformulating the NFA test once the tolerance threshold ϵ is set. More specifically,

$$\text{NFA}(I_{ij}) \leq \epsilon \iff \frac{m_1(I_{ij}) - m_2(I_{ij})}{\sigma} \geq \frac{\text{erfc}^{-1}\left(\frac{2\epsilon}{\#\mathcal{I}}\right)}{\sqrt{\frac{\pi}{2}}\sqrt{1 - \frac{1}{\alpha^2}} R}$$

2.2.2 Results over the synthetic data

The a-contrario particle detector is evaluated over the synthetic data of which the generation process was described in part 1.2. Since the data is generated, ground truth is accessible and the amount of noise can be controlled.

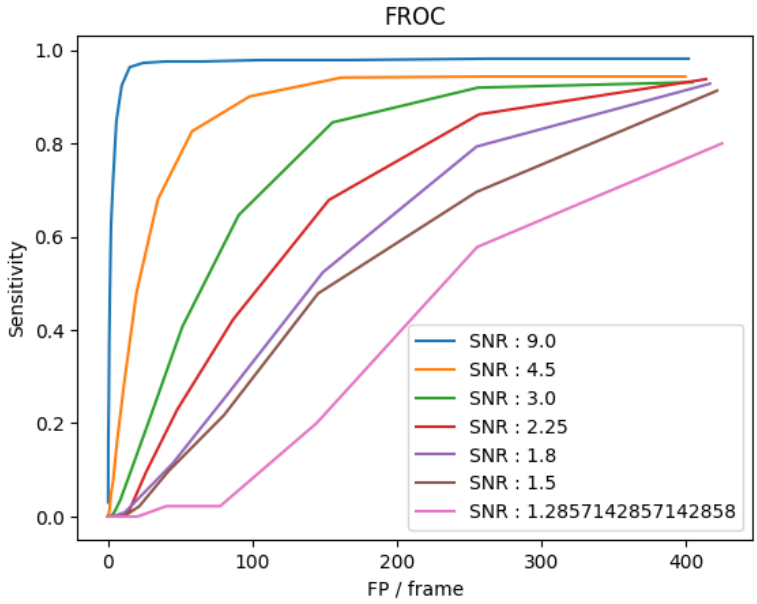

Figure 5: A-contrario particle detector evaluation over a synthetic particles simulation

Figure 5 shows the results of the a-contrario detector for different level of noise in the generated data. Two main observation can be made :

1. At high SNR, the detector is almost perfect. This high performance is not surprising since the synthetic data were generated under the detector's working assumption, which are a stationnary gaussian noise background and particles having the visual structure of a gaussian spot.

2. The performances decreases with the SNR. This phenomenon is also not surprising because at high SNR, the gaussian spots structure tend to blend in the background noise.

2.2.3 Results over more challenging data

The a-contrario detection method was also evaluated against a more challenging dataset. Since particles look like spots, the detection approach should generalize to any spot detection task. Based on that statement, the detector was evaluated on the Fluorescent Neuronal Cells dataset. This dataset contains images of mice brain cells obtained by fluorescent microscopy which should be located in the best way possible in order to assess the result of a biological experiment. For this evaluation experiment, vicinity grouping was not used since the goal is to perform segmentation and not point detection.

Figure 2.2.3 present a pathologic case and figure 8 present the best FROC curve that could be obtained after tuning R and α by a grid search.

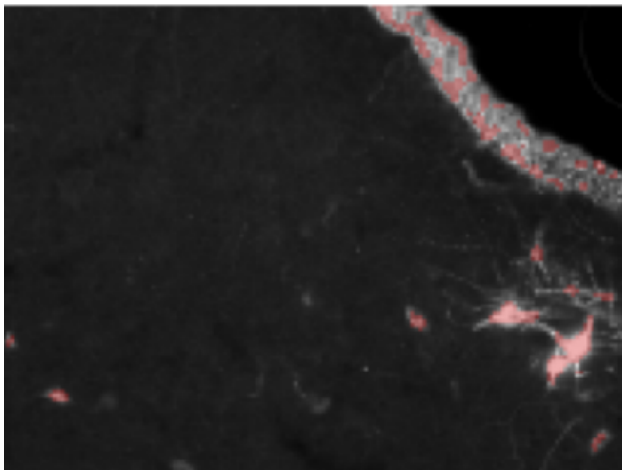

Figure 6: Prediction

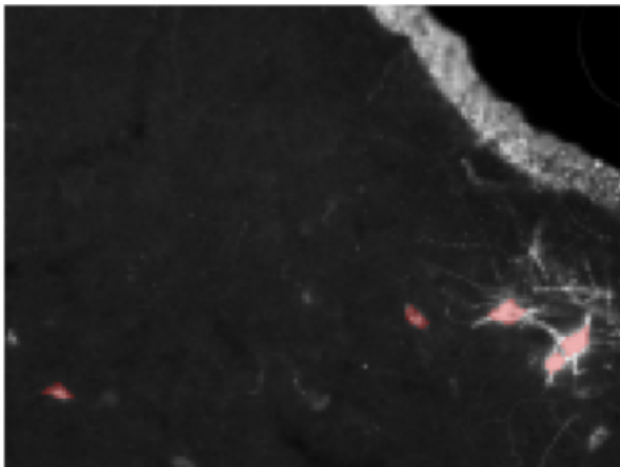

Figure 7: Ground truth

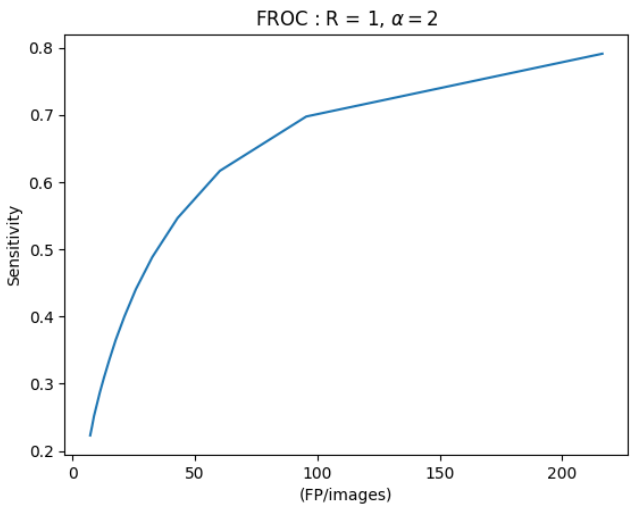

Figure 8: Caption

3 Particle Tracking

3.1 Methods

3.1.1 Background model

As before, the background hypothesis is that, when there is nothing to detect, the observed data is just sensor noise which is modelled by a gaussian stationnary noise. When the particle detector is applied on pure noise, false alarm appear which is translated to points that randomly appear on the prediction. Consequently, under pure noise in the input data, the intermediary data that is fed to the trajectory detector is points randomly appearing. Since, the noise is stationnary, points have equal probability of being misclassified. Thus, under the background hypothesis, the intermediary data is an image where the pixel are iid realization of a Bernoulli law $\mathcal{B}(p(\epsilon))$ where the rate depends on the tolerance ϵ of the particle detector.

3.1.2 Measurement Statistic

The detector aims at fining physically consistent trajectories. Such trajectories must be smooth, meaning that the maximal acceleration should remain reasonably low.

Based on this statement Primet and Moisan [2012] propose to measure the relevance of a trajectory by looking at its maximal acceleration. Since the actual data is a video with a discrete frame number,

the notion of acceleration is defined by a finite difference scheme. More formally,

- A trajectory τ consists of an ordered set of points such that consecutive points belongs to consecutive frames. Additionally, a trajectory must have at least three points. The length of a trajectory L is defined as its cardinality.
- The set of possible trajectory of length L that can be made from the data is noted $\mathbb{T}_L$ and the set of all possible trajectory is noted $\mathbb{T}$.
- For a given trajectory $\tau \in \mathbb{T}_L$, one defines its maximal acceleration as the quantity

$$a_{\max}(\tau) = \min_{i \in [|2, L-1]} \|\tau_{i+1} - 2\tau_i + \tau_{i-1}\|_2$$

However, the points that are consider live in a discrete finite grid. This makes the previous maximal acceleration definition unstable for trajectory extraction Primet and Moisan [2012]. In the case of a long trajectory having a sub-trajectory of length zero, the zero-acceleration chunk will be extremely meaningful which in turns will encourage the detector to split the long trajectory into distinct chunks to return the zero-acceleration sub-trajectory as one trajectory in itself. To mitigate this phenomena, the norm in the maximal acceleration definition is replaced by a pseudo-norm that cannot be zero-valued. The author propose to count the number of pixels contained in the closed discrete ball of radius $\|\tau_{i+1} - 2\tau_i + \tau_{i-1}\|_2$ and to normalize it with the number of pixel contains in a frame. This is encapsulated in the following formula

$$a^d_{\max}(\tau) = \min_{i \in [|2, L-1]} \frac{\#\mathbb{Z}^2 \bigcap \overline{\mathcal{B}}(0, \|\tau_{i+1} - 2\tau_i + \tau_{i-1}\|_2)}{\#\mathcal{I}}$$

3.1.3 NFA

Following the a-contrario paradigm, an NFA is built by weighting the probability of observing an unsmoothed trajectory under the background model. As opposed to the particle detection case, where no detection case was privileged, in the case of point tracking, small trajectories have higher chances of being detected. This is mitigated by a length dependent weighting. The standard NFA formulation that corresponds to this situation is the following :

$$\mathrm{NFA}(\tau) = w_L \mathbb{P}_{\mathcal{H}_0}\left(-a^d_{\max}(\tau) \leq -a^d_{\max}(\mathcal{T})\right), \quad \forall \tau \in \mathbb{T}_L$$

s.t

$$\sum_{L=3}^{N} \sum_{\tau \in \mathbb{T}_L} \frac{1}{w_L} = 1, \quad \text{where } N \text{ is the number of frames in the video}$$

The proposed NFA gives more significance to long trajectories thus balancing the chances of detecting each type of trajectory under the background model. This yields the following NFA :

$$\mathrm{NFA}(\tau) = K \cdot (K - L + 1) \cdot N^L \cdot (a^d_{\max}(\tau))^{L-2}, \quad \forall \tau \in \mathbb{T}_L$$

One can effectively verify that the weighting satisfies the NFA condition :

$$\sum_{\tau \in \mathbb{T}} \frac{1}{K \cdot (K - L + 1) \cdot N^L} = \sum_{L=3}^{N} \sum_{\tau \in \mathbb{T}_L} \frac{1}{K \cdot \#\mathbb{T}_L} = 1$$

3.2 Implementation

3.2.1 Implementation Details

A naïve way to perform the detection would be to evaluate every trajectory, resulting in an extremely complex algorithm. ASTRE was proposed by Primet and Moisan [2012] as a Bellman algorithm to extract meaningful trajectories without having to check every possibilities. However, ASTRE is still very slow in practice and could not run on the synthetic data. That is why CUTASTRE Abergel and Moisan [2014] was considered. This algorithm is an extension to ASTRE that breaks it into a frames chunks wise processing, drastically lowering the complexity of the method.

3.2.2 Results

In practice, only ASTRE could be implemented and the algorithm's performances did not match the one from the paper, mostly due to implementation issued that could not be fixed in time. On the one hand, the algorithm is extremelly slow and cannot be benchmarked or applied to a synthetic data in a reasonable amount of time. Nevertheless, a toy example of reduced size was implemented and a cherry picked result is provided figure 9

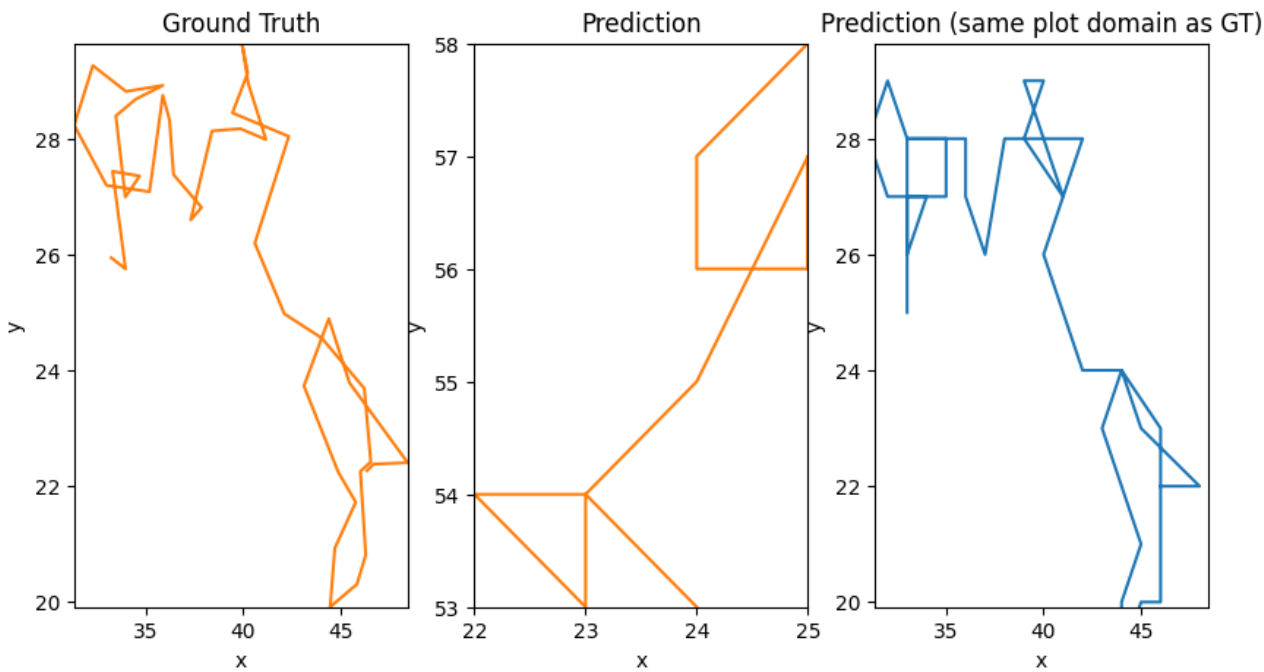

Figure 9: Output of our implementation of ASTRE

The result show that our implementation of ASTRE is subject to spurious detection with greatly break the consistency of the detected trajectory. However, zooming on the area of interest shows that the algorithm still capture a trajectory that locally resembles the ground truth.

Conclusion

This project aimed at reproducing the work of Dimiccoli et al. [2016] on particle detection and tracking. The method that was considered breaks down to two phase. First, an a-contrario particle detector is tasked of extracting points that might be identified as particle in each frame of a data video sample. This a-contrario detection relies on a residual noise background assumption paired with a gaussianity measurement statistic. The second phase aims at extracting trajectories from the point that were detected in phase 1. The method also relies on an a-contrario detection, where the background assumption is based on the image of the first phase's background model through the first phase's detection process. The detection statistic used in the second detection step identifies meaningful trajectories through their smoothness.

To make up for the unavailability of particle tracking dataset online, synthetic data was generated under mild assumptions. Then, these data were used to benchmark the first detection step. This benchmark showed that the approach could be reproduced with satisfying results. However, the second phase could not be reproduced, even if it was sometimes possible to recover encouraging cherry picked trajectory detection results.

A side evaluation was also conducted to evaluate how the first phase detector would perform under a broken background assumption. To that extent, it was evaluated over data with a textured background and surprisingly showed good results besides a significant amount of pathological cases due to the textured aspect of the background.

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